2019-Summer Paper 1 – Basic Sciences in Urology

Contents

1. Q1. Embryology and anatomy of the male urethra; endoscopic management of urethral strictures (LAQ)
2. Q2. Physiological and haemodynamic changes of pneumoperitoneum; complications of retroperitoneoscopy (LAQ)
3. Q3a. Methods to estimate GFR in a prospective renal donor (SAQ)
4. Q3b. VLPP in SUI (SAQ)
5. Q3c. HoLEP in BPH (SAQ)
6. Q3d. Non-obstructive azoospermia (SAQ)
7. Q3e. Acetylcysteine (SAQ)
8. Q3f. ESBL-producing enterobacterial UTI (SAQ)

Q1. Describe the embryology and anatomy of male Urethra. Describe different endoscopic management techniques of treatment of urethral strictures.

Type: LAQ (25 marks)

Embryology

- **Cloaca** is divided by the **urorectal septum** (weeks 4–7) into the anterior **UGS (urogenital sinus)** and the posterior anorectal canal.
- The UGS has three parts: **vesical** (bladder), **pelvic** (prostatic and membranous urethra) and **phallic** (urethral plate of the penis).
- **Posterior urethra:** from the pelvic UGS. Prostatic buds appear at 10 weeks; the **utricle** (Müllerian remnant) and ejaculatory ducts (Wolffian) open on the verumontanum.
- **Anterior urethra:** Leydig cells make testosterone from about 8 weeks; type 2 5α-reductase converts it to DHT (dihydrotestosterone), which masculinises the genital tubercle. Germ layers: ectoderm gives skin and prepuce, mesoderm the corpora, endoderm the **urethral plate**.
- **Penile urethra: "double zipper"**
  1. **Opening zipper:** the urethral plate canalises proximal to distal into a urethral groove between the urethral folds.
  2. **Closing zipper:** the folds fuse in the ventral midline, forming the tube and the median raphe. Folds fused by 16 weeks; urethra complete by about 18 weeks.

Glanular urethra	Ectodermal ingrowth theory (Glenister)	Endodermal theory (current)
Formed by	Surface ectoderm grows into the glans to meet the endodermal urethra	Direct canalisation of the urethral plate inside the glans
Epithelium	Mixed ectoderm and endoderm	Endoderm (UGS) along the whole urethra

- **Clinical link:** the shaft–glans junction (change of mechanism) is the commonest hypospadias site. **Also know:** urethral duplication; PUV (posterior urethral valves); utricular cyst.

Anatomy

- About 18–20 cm long in adults; normal diameter 8–9 mm. Posterior urethra runs from the bladder neck to the **perineal membrane**; anterior urethra from there to the meatus.

Segment	Length / features	Lining
Prostatic	Length of prostate; urethral crest, verumontanum, utricle, ejaculatory ducts	Urothelium
Membranous	2–2.5 cm; shortest and narrowest; striated sphincter	Pseudostratified columnar
Bulbar	Widens as the bulb; Cowper ducts at 3 and 9 o'clock; **eccentric, close to the dorsum** of spongiosum	Pseudostratified columnar
Penile	About 15 cm; central, uniform calibre; glands of Littré	Pseudostratified columnar
Fossa navicularis	Glanular; lacuna magna on the roof	Stratified squamous

- **Sphincters:** (1) **internal (preprostatic) smooth muscle sphincter** at the bladder neck, autonomic; (2) **external striated rhabdosphincter**, horseshoe-shaped around the membranous urethra, pudendal nerve (mainly S3, some S2; Onuf nucleus), plus its intrinsic smooth muscle.
- **Blood supply:** internal pudendal → common penile artery → artery to the bulb (proximal) and **retrograde flow via the dorsal arteries through the glans**. This dual supply allows bulbar transection in anastomotic urethroplasty.
- **Venous** to pudendal and prostatic plexuses; **lymph** to internal and external iliac nodes.

Endoscopic management of stricture

Principle: healing after incision or dilation is by **secondary intention**, a race between epithelialisation and wound contraction. No endoscopic method removes **spongiofibrosis**, so cure is limited to short, superficial bulbar strictures.

Favourable	Unfavourable
Bulbar, **<1.5–2 cm**	Penile, fossa, longer
No dense spongiofibrosis	Straddle injury, LS (lichen sclerosus)
First procedure	Repeat; luminal narrowing >74%

1. Dilation

- Stretches the scar without tearing it (bleeding means tearing).
- **Safest: balloon dilation over a guidewire** under vision; blind filiforms condemned. Sounds or self-dilation for palliation.

2. DVIU (direct vision internal urethrotomy)

1. RUG (retrograde urethrogram) and VCUG (voiding cystourethrogram) define site and length.
2. Guidewire across; **normal saline** irrigation.
3. **Sachse cold knife** at 12 o'clock, through scar to healthy tissue; avoid deep cuts (corpora: ED (erectile dysfunction), fistula).
4. Catheter **3–7 days**; longer adds nothing.

- **Laser urethrotomy** (holmium, others): **no advantage over cold knife**.
- Results: **Steenkamp and Heyns** (randomised): dilation equals urethrotomy; a second attempt rarely and a third never succeeds. **Santucci (2010):** about 8% stricture-free long term.

3. Adjuncts against recurrence

- **CISC (clean intermittent self-catheterisation)** tapering over 3–6 months: delays recurrence, which returns on stopping.
- **Mitomycin C** injection reduces recurrence; steroid injection, colchicine: weak data.

4. Urethral stents

- **UroLume** (permanent): **withdrawn** (ingrowth). **Memokath**, removable stents: small series only.

5. Optilume (paclitaxel-coated balloon)

- FDA (Food and Drug Administration) 2023. **ROBUST III:** 74.6% vs 26.8% stricture-free at 1 year after failed endoscopy.

Complications: recurrence (commonest), bleeding with erections, irrigant extravasation, UTI (urinary tract infection) and sepsis, ED, spongio-cavernous fistula, incontinence.

Guideline (AUA (American Urological Association) 2016): urethrotomy or dilation for short (<2 cm) bulbar strictures; **urethroplasty, not repeat endoscopy**, after failure. The reconstructive ladder is abandoned.

Diagram to draw: sagittal male urethra with segments, sphincters and a 12 o'clock urethrotomy.

> **Key points to remember:**
>
> - Double zipper: canalisation, then fold fusion; glanular urethra endodermal.
> - Bulbar urethra eccentric (dorsal); dual retrograde glanular blood supply.
> - DVIU: bulbar, <1.5–2 cm, first attempt; Santucci about 8% long-term.
> - Optilume ROBUST III 74.6% vs 26.8%.

Q2. Describe physiological and hemodynamic changes due to creation of pneumoperitoneum during laparoscopy. Describe complications of Retroperitoneoscopy.

Type: LAQ (25 marks)

Introduction

- **Pneumoperitoneum** is insufflation of gas, usually CO_2 (carbon dioxide), to 10–15 mmHg to create working space.
- Its effects have **two drivers**: **pressure** (compression of cava, kidneys, diaphragm) and **absorbed CO_2** (acidosis, sympathetic drive). Position modifies both.
- **CO_2** is standard: non-combustible, cheap, highly soluble (embolism less lethal). Helium or argon in severe lung disease; N_2O (nitrous oxide) supports combustion.

Effects by pressure (supine)

Effect	10 mmHg	20 mmHg	40 mmHg
Heart rate, MAP (mean arterial pressure), SVR (systemic vascular resistance)	↑	↑	MAP, SVR ↑; HR ↓
Venous return, cardiac output	Variable	Variable	↓ sharply
GFR (glomerular filtration rate), urine	↓	↓↓	↓↓
Arterial PCO_2 / pH	↑ / ↔	↑ / ↓	↑ / ↓

Cardiovascular

- **Afterload rises** at every pressure (aortic compression, catecholamines, vasopressin).
- **Preload:** in hypovolaemia the cava collapses and venous return falls; in hypervolaemia it may rise. Above 40 mmHg output collapses.
- **Arrhythmia:** hypercapnia causes tachycardia and ectopics; peritoneal stretch causes **vagal bradycardia** or asystole. Arrhythmia may herald pneumothorax, hypoxia or embolism.
- **CVP (central venous pressure) is falsely high** (raised intrathoracic pressure).
- Leg venous stasis: theoretical DVT (deep vein thrombosis) risk.

Respiratory

- Diaphragm splinted: **FRC (functional residual capacity) and compliance fall**, peak airway pressure rises; atelectasis, V/Q mismatch. PEEP (positive end-expiratory pressure) helps.
- Steep Trendelenburg adds endobronchial tube slip and facial oedema.

Acid–base (CO_2 absorption)

- CO_2 is absorbed across the peritoneum (and through fat and lymphatics extraperitoneally); rises over 30–60 minutes, then plateaus.
- **Hypercarbia and respiratory acidosis**; managed by raising minute ventilation and lowering pressure.
- **COPD (chronic obstructive pulmonary disease)** patients cannot blow it off: arterial gases if >1 hour; consider helium. Stored CO_2 causes **peak hypercarbia after extubation**.

Renal

- **Oliguria** from **renal vein and parenchymal compression** (not ureteric or hormonal); cortical flow falls.
- Keep pressure ≤10–12 mmHg; avoid fluid overload; mannitol or furosemide if needed. Reversible.

Other systems

- **Splanchnic** flow falls (rare mesenteric thrombosis); less ileus than open surgery.
- **ICP (intracranial pressure)** and **IOP (intraocular pressure)** rise with hypercarbia and head-down tilt: caution in head injury, shunts, glaucoma.
- **Hormonal:** catecholamines, cortisol and **vasopressin** rise; renin–angiotensin activation adds to afterload.
- **Stress and immune response** smaller than open surgery (less hyperglycaemia and catabolism), which may explain faster recovery.
- Shoulder-tip pain from diaphragmatic CO_2 irritation; hypothermia from dry cold gas.

Position effects

Position	Effect
Head-down	Venous return and output ↑; lung volumes ↓, ICP and IOP ↑
Head-up	Output ↓, hypotension
Lateral flank	Lung V/Q mismatch; rhabdomyolysis with kidney rest

- **Practical rule:** work at 10–12 mmHg; never exceed 20 mmHg for long. Retroperitoneal insufflation causes smaller haemodynamic swings.

Complications of retroperitoneoscopy

Why different: small space with no natural cavity, created by finger and **balloon dissection** (commercial or Gaur glove balloon) behind Gerota's fascia, in front of psoas; fewer landmarks; gas tracks along tissue planes; CO_2 absorbed through fat and disrupted lymphatics, so hypercarbia appears early if the port leaks.

1. Access and balloon-related

- Balloon in the wrong plane (intramuscular → hernia); **balloon rupture** (latex fragments).
- **Peritoneal tear:** CO_2 enters the peritoneum and collapses the working space. Fix by an extra port to vent, or convert to transperitoneal.

2. Gas-related

- **Subcutaneous emphysema** (commonest): crepitus over flank, chest and neck, pneumoscrotum; prevent with a **balloon-tip port**, tight incisions, 12 rather than 15 mmHg.
- **Hypercarbia**, especially with emphysema.
- **Pneumothorax** (1.6–4%, commoner retroperitoneally), pneumomediastinum, pneumopericardium: needle decompression, chest drain.
- **Gas embolism** (renal vein injury): collapse, end-tidal CO_2 rises then falls, mill-wheel murmur. Stop gas, **left lateral head-down**, 100% oxygen, aspirate through a central line.

3. Vascular injury

- Commoner in upper-tract (retroperitoneal) work: renal vessels, IVC (inferior vena cava), lumbar veins, gonadal vessels, aorta.
- Increase pressure temporarily, compress, add ports, suture or clip; convert if uncontrolled.

4. Visceral injury

- Pleura, peritoneum, colon, duodenum, pancreas, spleen, liver, ureter; usually by thermal spread.

5. Postoperative

- Port-site hernia or haematoma; flank bulge; neuropraxia (lateral femoral cutaneous, genitofemoral); **rhabdomyolysis** (about 1%; limit kidney rest); urinoma, lymphocele; delayed bowel injury; conversion to open or transperitoneal surgery.

Diagram to draw: flank port positions for retroperitoneoscopy with balloon in the space between psoas and Gerota's fascia.

> **Key points to remember:**
>
> - Two drivers: pressure and CO_2; work at 10–12 mmHg.
> - Afterload rises; oliguria from renal vein compression; peak hypercarbia after extubation.
> - Gas embolism: left lateral, head down, oxygen, aspirate.
> - Retroperitoneoscopy: emphysema and peritoneal tear; balloon-tip port prevents leak.

Q3a. Describe methods to estimate GFR in a prospective renal donor

Type: SAQ (10 marks)

Why it matters

- Donor loses about half the nephron mass. **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** sets GFR (glomerular filtration rate) thresholds and asks for **measurement, not just estimation**, at a decision point.

Methods

1. Serum creatinine and eGFR (estimated GFR)

- Creatinine is freely filtered, but generation depends on **muscle mass, diet and age**; it is also secreted, so it overestimates GFR.
- **CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) 2021 creatinine equation:** age and sex, **no race term**; currently recommended. MDRD (Modification of Diet in Renal Disease) and Cockcroft–Gault are older.
- Less accurate at GFR >60, where donors lie: a screening test only.

2. Cystatin C

- 13-kDa protein made by all nucleated cells; freely filtered, fully reabsorbed and broken down. Independent of muscle mass; affected by thyroid state, steroids, inflammation.
- **Combined creatinine–cystatin C equation** is more accurate than either alone.

3. Measured 24-hour creatinine clearance

- (Urine creatinine × volume) ÷ serum creatinine. Prone to collection errors; secretion overestimates GFR.

4. mGFR (measured GFR): reference

Marker	Method
Inulin	Urinary clearance; gold standard, research only
Iohexol	Plasma clearance; safe, accurate
Iothalamate	Urinary or plasma clearance
^{51}Cr-EDTA (ethylenediaminetetraacetic acid)	Plasma clearance
^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid)	Plasma sampling, or gamma camera (Gates method)

5. Split function

- DTPA renogram gives differential function; MAG3 or DMSA alternatives. A difference >10% (beyond 45:55) matters: **leave the better kidney with the donor**.

KDIGO 2017 approach

- Initial **eGFRcr**; confirm with **mGFR, measured creatinine clearance or eGFRcr-cys**.

GFR (mL/min/1.73 m²)	Decision
≥90	Accept
60–89	Individualise (age, lifetime risk)
<60	Decline

- Indian practice: **DTPA GFR** with split function plus CT (computed tomography) angiography is standard.

Diagram to draw: plasma tracer clearance curve after a single injection.

Key points to remember:

- CKD-EPI 2021 race-free; cystatin C adds accuracy.
- Inulin is gold standard; iohexol or DTPA in practice.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.
- Better kidney stays with the donor.

Q3b. VLPP in SUI

Type: SAQ (10 marks)

Definition

- **VLPP (Valsalva leak point pressure):** the **intravesical pressure** at which urine leaks during Valsalva, **without a detrusor contraction**.
- It is a type of **ALPP (abdominal leak point pressure)**, which measures the urethra's ability to resist abdominal pressure. Lower value = weaker sphincter.
- Seen only in SUI (stress urinary incontinence); **no normal value** (continent women never leak).

Technique

1. Small dual-lumen urethral catheter plus rectal catheter; zero to atmosphere.
2. Patient **standing or sitting**; fill to **150 mL** (original method; some use 200 mL).
3. Ask for a **graded Valsalva**; record the **lowest total Pves (vesical pressure)** at visible leak (include baseline, not the change).
4. No leak: repeat every 50 mL to capacity; then **remove the urethral catheter** and read pressure from the rectal line.
5. Report volume, catheter size and provocation (Valsalva or cough).

Interpretation (McGuire 1993)

VLPP (cmH_2O)	Inference
<60	**ISD (intrinsic sphincter deficiency)**
60–90	Equivocal; mixed
>90	**Urethral hypermobility**, little ISD

Limitations

- Not standardised: falls with **larger catheters** and **higher volumes**.
- **VLPP is lower than CLPP (cough leak point pressure)** in the same woman; state the method.
- Invalid during DO (detrusor overactivity) or poor compliance; prolapse raises it (reduce it).
- ISD and hypermobility coexist; VLPP **does not predict surgical outcome**.
- **VALUE trial (Nager 2012):** for uncomplicated SUI, office evaluation was non-inferior to urodynamics before surgery. AUA/SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2023: urodynamics optional in the index patient.

VLPP vs DLPP

VLPP	DLPP (detrusor leak point pressure)
Sphincter vs abdominal pressure	Bladder pressure vs outlet resistance
Low = bad	**High (>40 cmH_2O) = bad**: upper-tract risk (McGuire 1981)
SUI	Neurogenic, low-compliance bladder

Also know: MUCP (maximum urethral closure pressure) ≤20 cmH_2O suggests ISD.

Diagram to draw: Pves, Pabd, Pdet tracing with leak marked on Valsalva.

Key points to remember:

- Pves at leak with Valsalva, no detrusor contraction; 150 mL.
- <60 ISD; 60–90 equivocal; >90 hypermobility.
- VLPP < CLPP; catheter size and volume alter it.
- DLPP >40: upper-tract danger.

Q3c. HoLEP in BPH

Type: SAQ (10 marks)

Principle

- **HoLEP (holmium laser enucleation of the prostate)** removes the whole adenoma along the **surgical capsule**, as in open simple prostatectomy, but endoscopically.
- **Ho:YAG (holmium:yttrium–aluminium–garnet), 2120 nm**: absorbed by water, pulsed, **0.4 mm penetration**, coagulates vessels. Normal saline irrigation (no TUR (transurethral resection) syndrome).
- Evolution: HoLAP → HoLRP → **HoLEP (1998)**, made practical by the **morcellator**.

Indications

- **Size-independent** (<30, 30–80, >80 mL); median lobe; **anticoagulated or high bleeding risk** (AUA (American Urological Association) recommendation); alternative to open prostatectomy for large glands.

Steps

- 60–150 W laser, 550 µm end-firing fibre, 24–28 Fr resectoscope; about 2 J × 40–50 Hz to enucleate.

1. Bladder-neck incisions at 5 and 7 o'clock to capsule (median lobe) or one at 6 o'clock.
2. Perimontanal incision; **early apical release** to spare the sphincter.
3. Develop the capsular plane round to 12 o'clock; detach lobes into the bladder (**en bloc, 2-lobe or 3-lobe**).
4. Haemostasis with a defocused beam.
5. **Morcellation** with a full bladder; 22 Fr three-way catheter.

Outcomes

- Versus **TURP (transurethral resection of the prostate)** (trials, meta-analyses): **better IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate)**, less bleeding, shorter catheter time and stay; longer operation. The **only endoscopic method superior to TURP**.
- Versus **open prostatectomy** for large glands (**Kuntz, 5-year**): equal function with less blood loss and shorter stay.
- **Durable**: whole transition zone removed; **retreatment rare**.

Complications

- **Retrograde ejaculation** about 75%; transient storage symptoms and early SUI (stress urinary incontinence).
- BNC (bladder neck contracture) 0–3.2%; stricture; capsular perforation; ureteric orifice injury; **morcellator bladder injury**; retained adenoma.
- Erectile function unchanged.

Limitation

- **Steep learning curve**: efficiency plateaus after about 50 cases.

Recent advances

- **Moses pulse modulation**; thulium enucleation; ejaculation-sparing HoLEP; same-day discharge. EAU (European Association of Urology): enucleation is an alternative to TURP, and to open surgery for >80 mL.

Diagram to draw: sagittal prostate showing enucleation along the capsule.

Key points to remember:

- Ho:YAG 2120 nm, water chromophore, 0.4 mm.
- Size-independent; best for anticoagulated men.
- Better IPSS and Qmax than TURP; durable; retreatment rare.
- Retrograde ejaculation about 75%; learning curve about 50 cases.

Q3d. Non obstructive azoospermia

Type: SAQ (10 marks)

Definition

- **NOA (non-obstructive azoospermia):** no sperm in the **centrifuged** ejaculate on two samples, due to **failure of spermatogenesis** (testicular failure). About 60% of azoospermic men.

Causes

- **Genetic: Klinefelter (47,XXY)**, the commonest; **Y microdeletion** (AZF (azoospermia factor) a, b, c); other karyotype anomalies.
- **Acquired:** cryptorchidism, mumps orchitis, torsion, **chemotherapy or radiation**, trauma, varicocele.
- **Idiopathic**: commonest overall.
- *Note:* hypogonadotropic hypogonadism is pre-testicular and treatable medically (hCG (human chorionic gonadotropin) ± FSH (follicle-stimulating hormone)).

Evaluation

- Small soft testes, palpable vasa, normal ejaculate volume.
- **FSH >7.6 IU/L and testis long axis ≤4.6 cm** point to NOA (FSH ≤7.6 with normal testes suggests obstruction).
- LH (luteinising hormone), testosterone, oestradiol, prolactin.
- **Genetics (AUA/ASRM (American Urological Association/American Society for Reproductive Medicine)):** karyotype and Y microdeletion in all NOA; counselling.
- **Histology patterns:**

Pattern	Meaning	Micro-TESE retrieval
Hypospermatogenesis	All stages, fewer cells	Best
Maturation arrest	Halts at a germ-cell stage	About 40%
SCO (Sertoli cell–only)	No germ cells	About 30%

Management

1. Optimisation (before retrieval)

- Stop exogenous testosterone; repair clinical varicocele (no definitive evidence in NOA).
- Hormonal therapy off-label: clomiphene, anastrozole, hCG for low testosterone.

2. Micro-TESE (microdissection testicular sperm extraction)

- **Principle:** spermatogenesis in NOA is **focal**. Under ×20–25 magnification, **larger, opaque tubules** are chosen; least tissue removed, blood supply preserved.
- Sperm found in **about 50%**; better than conventional TESE.

Predictor	Retrieval (%)
Post-orchidopexy	74
AZFc deletion	70
Klinefelter	68
Post-chemotherapy	53
AZFa or AZFb complete deletion	**0: do not operate**

- FSH and testis size **do not** reliably predict retrieval.

3. ICSI (intracytoplasmic sperm injection)

- Fresh or frozen testicular sperm. Live birth per cycle modest; miscarriage higher (sperm DNA damage). Male offspring of AZF deletion carriers inherit it.

4. No sperm: donor sperm or adoption.

Also know: in-vitro spermatogenesis and stem-cell transplantation are experimental.

Diagram to draw: opened testis under microscope: plump tubules versus thin Sertoli cell–only tubules.

Key points to remember:

- FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in all.
- Micro-TESE: sperm in about half; SCO 30%.
- AZFa/b deletion: no retrieval.

Q3e. Acetylcysteine

Type: SAQ (10 marks)

Answered in 2018-Summer Paper 1, Q3b; core answer repeated here.

Introduction

- **NAC (N-acetylcysteine):** acetylated cysteine, a **thiol (–SH) compound**; oral, intravenous or nebulised.

Mechanisms of action

- **Glutathione precursor:** supplies cysteine, the rate-limiting substrate for **glutathione**.
- **Antioxidant:** the free –SH group scavenges reactive oxygen species.
- **Mucolytic:** breaks **disulfide bonds** in mucoproteins.
- **Thiol–disulfide exchange:** in theory forms a soluble mixed disulfide with cystine.

Clinical uses

1. Paracetamol (acetaminophen) overdose: main indication

- Overdose saturates normal pathways; the toxic metabolite **NAPQI (N-acetyl-p-benzoquinone imine)** builds up once glutathione is exhausted, causing liver necrosis.
- NAC restores glutathione; most effective **within 8 hours**, but still given later.
- **IV (intravenous) 21-hour regimen:** 150 mg/kg, then 50 mg/kg over 4 h, then 100 mg/kg over 16 h. Shorter two-bag regimens are used in some countries.

2. Mucolytic: COPD (chronic obstructive pulmonary disease), cystic fibrosis.

Urological relevance

- **Contrast-induced AKI (acute kidney injury):** once widely used. **PRESERVE (2018):** no benefit over saline. **No longer recommended**; isotonic hydration is the proven measure.
- **Cystinuria:** thiol action theoretical; standard thiols are **tiopronin** and D-penicillamine.
- **Cyclophosphamide cystitis: mesna** (another thiol) is used.
- **Mucus after enterocystoplasty or neobladder:** oral or irrigation; weak evidence.
- **Male infertility:** antioxidant for oxidative stress (high ROS (reactive oxygen species), sperm DNA fragmentation); limited evidence.
- **Ischaemia–reperfusion** (renal surgery, transplant): antioxidant protection studied; not established.

Adverse effects

- **Anaphylactoid reactions** with IV loading (flushing, rash, bronchospasm): slow the infusion, antihistamine.
- Nausea, vomiting, sulphur smell; bronchospasm when nebulised in asthma.
- Safe in pregnancy and renal impairment for paracetamol poisoning; benefit outweighs risk.

Diagram to draw: paracetamol → NAPQI → glutathione conjugation, with NAC replenishing glutathione.

> **Key points to remember:**
>
> - Thiol; glutathione precursor, antioxidant, mucolytic.
> - Antidote for paracetamol overdose; best within 8 h.
> - PRESERVE 2018: no benefit in contrast AKI.
> - Cystinuria uses tiopronin, not NAC.

Q3f. ESBL producing enterobacterial UTI

Type: SAQ (10 marks)

Definition

- **ESBLs (extended-spectrum β-lactamases): plasmid-encoded** enzymes that hydrolyse penicillins, **third-generation cephalosporins** and aztreonam. They are **inhibited by clavulanate** and spare **carbapenems** and cephamycins.
- Mainly *E. coli* and *Klebsiella* (Enterobacterales); first reported 1983.

Mechanism and spread

- Older types **TEM** and **SHV**; now **CTX-M** dominates worldwide. Spread through the community is linked to *E. coli* **ST131** (fluoroquinolone resistance is its hallmark).
- Plasmids carry co-resistance genes: **fluoroquinolones, aminoglycosides, TMP-SMX (trimethoprim–sulfamethoxazole)**. So oral options are few.
- India has one of the highest prevalence rates.

Risk factors

- Prior antibiotics (especially cephalosporins, quinolones); recent hospital stay; **indwelling catheter**, urological instrumentation; recurrent UTI; diabetes; nursing-home residence; travel to high-prevalence areas.

Detection

- Screening: reduced susceptibility to ceftriaxone, cefotaxime or ceftazidime.
- **Phenotypic confirmation:** clavulanate synergy (**combination disc** or double-disc synergy test); automated systems.
- Genotypic PCR (polymerase chain reaction) for CTX-M.

Treatment

Infection	Options
Cystitis	**Nitrofurantoin** 5 days, **fosfomycin** 3 g single dose (E. coli), **pivmecillinam**; TMP-SMX or quinolone if susceptible; single-dose aminoglycoside
Pyelonephritis, complicated UTI	Quinolone or TMP-SMX if susceptible; else **aminoglycoside**
Urosepsis, bacteraemia	**Carbapenem** (meropenem, ertapenem), then oral step-down by culture

- **MERINO trial (2018):** in ceftriaxone-resistant *E. coli*/*Klebsiella* bacteraemia, **piperacillin–tazobactam** had higher 30-day mortality than meropenem (12.3% vs 3.7%). Avoid pip-taz for serious ESBL infection.
- **Ceftolozane–tazobactam, ceftazidime–avibactam:** active but **reserved** for carbapenem-resistant organisms (stewardship).
- Remove or change catheters; drain obstruction (source control).
- Treat **asymptomatic bacteriuria** only before urological procedures or in pregnancy.

Prevention

- **Antibiotic stewardship:** avoid broad-spectrum cephalosporins and quinolones; shortest effective course; culture before treatment.
- Hand hygiene, contact precautions, catheter care.

- Pre-biopsy rectal swab culture for targeted prophylaxis; transperineal biopsy.

Also know: AmpC (not clavulanate-inhibited) and **KPC (*Klebsiella pneumoniae* carbapenemase)**, NDM (New Delhi metallo-β-lactamase) are wider resistance.

Diagram to draw: treatment flowchart from culture result to oral or IV choice.

Key points to remember:

- Plasmid CTX-M; resists 3rd-gen cephalosporins; inhibited by clavulanate.
- Oral: nitrofurantoin, fosfomycin, pivmecillinam.
- Serious infection: carbapenem; MERINO: pip-taz inferior.
- Stewardship prevents spread.

Rapid Revision Sheet – 2019-Summer Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe the embryology and anatomy of male Urethra. Describe different endoscopic management techniques of treatment of urethral strictures. (LAQ)

- Double zipper: canalisation, then fold fusion; glanular urethra endodermal.
- Bulbar urethra eccentric (dorsal); dual retrograde glanular blood supply.
- DVIU: bulbar, <1.5–2 cm, first attempt; Santucci about 8% long-term.
- Optilume ROBUST III 74.6% vs 26.8%.

Q2. Describe physiological and hemodynamic changes due to creation of pneumoperitoneum during laparoscopy. Describe complications of Retroperitoneoscopy. (LAQ)

- Two drivers: pressure and CO_2; work at 10–12 mmHg.
- Afterload rises; oliguria from renal vein compression; peak hypercarbia after extubation.
- Gas embolism: left lateral, head down, oxygen, aspirate.
- Retroperitoneoscopy: emphysema and peritoneal tear; balloon-tip port prevents leak.

Q3a. Describe methods to estimate GFR in a prospective renal donor (SAQ)

- CKD-EPI 2021 race-free; cystatin C adds accuracy.
- Inulin is gold standard; iohexol or DTPA in practice.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.
- Better kidney stays with the donor.

Q3b. VLPP in SUI (SAQ)

- Pves at leak with Valsalva, no detrusor contraction; 150 mL.
- <60 ISD; 60–90 equivocal; >90 hypermobility.
- VLPP < CLPP; catheter size and volume alter it.
- DLPP >40: upper-tract danger.

Q3c. HoLEP in BPH (SAQ)

- Ho:YAG 2120 nm, water chromophore, 0.4 mm.
- Size-independent; best for anticoagulated men.
- Better IPSS and Qmax than TURP; durable; retreatment rare.
- Retrograde ejaculation about 75%; learning curve about 50 cases.

Q3d. Non obstructive azoospermia (SAQ)

- FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in all.
- Micro-TESE: sperm in about half; SCO 30%.
- AZFa/b deletion: no retrieval.

Q3e. Acetylcysteine (SAQ)

- Thiol; glutathione precursor, antioxidant, mucolytic.
- Antidote for paracetamol overdose; best within 8 h.
- PRESERVE 2018: no benefit in contrast AKI.
- Cystinuria uses tiopronin, not NAC.

Q3f. ESBL producing enterobacterial UTI (SAQ)

- Plasmid CTX-M; resists 3rd-gen cephalosporins; inhibited by clavulanate.
- Oral: nitrofurantoin, fosfomycin, pivmecillinam.

- Serious infection: carbapenem; MERINO: pip-taz inferior.
- Stewardship prevents spread.